NOT MEASUREMENT SENSITIVE

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SUPERSEDING
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MILITARY SPECIFICATION

SEMITRAILER, VAN; AIR TRANSPORTABLE, 4990 KILOGRAMS (5.5 TON), AIR CONDITIONED, MODIFIED COMMERCIAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 4990 kilogram (kg) (5.5 ton) capacity semitrailer equipped with a demountable axle suspension system to enable aircraft loading. The semitrailer is air conditioned and equipped with a single axle. Semitrailers procured under this specification are modified commercial items which are required to be warranted by the manufacturer as specified in acquisition documents.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS), and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC-2330

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SPECIFICATIONS
FEDERAL

*	A-A-393	- Extinguishers, Fire, Dry-Chemical, (Hand-Portable).
	L-F-475	- Floor Covering, Vinyl, Surface (Tile and Roll) With Backing.

MILITARY

	MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.

STANDARDS
FEDERAL

	FED-STD-297	- Rustproofing of Commercial (Nontactical) Vehicles.

MILITARY

	MIL-STD-171	- Finishing of Metal and Wood Surfaces.
*	MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
	MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plate Standards.
*	MIL-STD-1595	- Qualification of Aircraft, Missile and Aerospace Fusion Welders.
	MIL-STD-1791	- Designing for Internal Aerial Delivery in Fixed Wing Aircraft.

(Application for copies of MIL-STD-1791 should be addressed to Wright-Patterson AFB, ASD/ENESS, Mr. Scott Kuhnen, Wright-Patterson, AFB, OH 45433, telephone (513)255-6281.)

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF DEFENSE
Department of Defense Index of Specifications and Standards (DODISS).

(Copies of the DODISS are available on a yearly subscription basis either from the Government Printing Office for hard copy, or microfiche copies are available from the Director, Navy Publication and printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093.)

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DEPARTMENT OF TRANSPORTATION (DoT)
Federal Motor Carrier Safety Regulations (FMCSR).
Federal Motor Vehicle Safety Standards (FMVSS).

* (Application for copies of DOT publications should reference the Code of Federal Regulations, 49 CFR, and the Federal Register, and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

AFSC Design Handbook
DH-1-11 - Air Transportability Handbook.

(Application for copies of the AFSC Design Handbook should be addressed to Wright-Patterson AFB, ASD/ENESS, Mr. Rod Wright, Wright-Patterson AFB, OH 45433, telephone (513)255-6281.)

* OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)
Code of Federal Regulations, 29 CFR.

(Application for copies of OSHA publications should reference the Code of Federal Regulations, 29 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are issues of the documents cited in the solicitation (see 6.2).

THE AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
ASME Boiler and Pressure Vessel Code, Section VIII.

(Application for copies of ASME publications should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East Forty-Seventh Street, New York, NY 10017.)

* AMERICAN WELDING SOCIETY (AWS)
AWS B2.1 - Standard for Welding Procedure and Performance Qualifications.
AWS D1.1 - Structural Welding Code-Steel.

(Application for copies of AWS publications should be addressed to the American Welding Society, 2501 NW 7th Street, Miami, FL 31025.)

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THE EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION (ETRTO)
Standards Manual.

(Application for copies of the ETRTO publications should be addressed to the European Tyre and Rim Technical Organisation, 32, Avenue Brugmann, 1060 Brussels, Belgium.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

*	NFPA 10	- Portable Fire Extinguisher.
	NFPA 70	- National Electrical Code.

* (Application for copies of NFPA publications should be addressed to National Fire Protection Association, One Battery March Park, Quincy, MA 02269-9101.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)
SAE Standards and Recommended Practices

J318	- Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers - Trucks, Truck-Tractors, and Trailers (DOD adopted).
J478	- Slotted and Recessed Head Screws.
J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (30-1000 MHz).
J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
J682	- Rear Wheel Splash and Stone Throw Protection (DOD adopted).
J700	- Upper Coupler Kingpin - Commercial Trailers and Semitrailers.
J702	- Brake and Electrical Connection Locations - Truck-Tractor and Truck-Trailer.
J1292	- Automobile, Truck, Truck Tractor, Trailer, and Motor Coach Wiring.

(Application for copies of SAE publications should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Year Book.

* (Application for copies of Tire and Rim Association publications should be addressed to the Tire and Rim Association, Inc., 175 Montrose West Avenue, Copley, OH 44321.)

* (Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Standard semitrailer, components, and accessories. Except as specified in 3.1.1 through 3.1.1.13, the semitrailer, components, and accessories to be delivered under the contract shall be the manufacturer's standard or optional items, sold commercially to the public, which meet or exceed the requirements of this specification. All items shall be as represented and rated in the manufacturer's technical data, including special or mounted equipment. Technical data shall be limited to specifications and technical material identical to that furnished to the authorized company representatives for selection of semitrailer models and components, and shall be available to the engineering offices of the procuring activity prior to delivery of the items.

3.1.1 Special requirements. In addition to the standard semitrailer and components specified in 3.1, the semitrailer shall be furnished with special equipment as specified herein.

* 3.1.1.1 Treatment and painting. The vehicle shall be treated and painted in accordance with MIL-STD-1223. As specified by the procuring activity for the appropriate military service (see 6.2), the exterior color shall be in accordance with MIL-STD-1223. The body interior color shall be the manufacturer's standard gloss white.

* 3.1.1.2 Marking and data plates. As specified by the procuring activity for the appropriate military service (see 6.2), identification marking and data plates shall be in accordance with MIL-STD-1223.

* 3.1.1.2.1 Instruction plates. Operating instruction plates for at least the following equipment shall be furnished:

(a) Electrical circuits controlled by switches
(b) Air line outlets
(c) Air conditioning controls
(d) Heating controls
(e) Schematic of interior electrical circuits
(f) Schematic of air lines
(g) Complete operating instructions for preparing the vehicle for air transport.

Instruction plates, schematics, and control devices shall be depicted on photoetched, corrosion-resistant plates. Control device plates shall be secured near the control. Instruction and schematic plates shall be located

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on the curbside wall near the electrical distribution boxes. Operating instructions for preparing the vehicle for air transport shall be located near or on the interior of the curbside door.

* 3.1.1.3 Rustproofing. The vehicle shall be rustproofed in accordance with FED-STD-297.

* 3.1.1.4 Brake lights. At least one pair of brake lights shall override the four-way emergency flasher or the two systems shall be independent of each other. Modifications to the manufacturer's standard product to accommodate this requirement shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard. If additional lights are added to the vehicle, the lights shall be selected from the semitrailer manufacturer's standard matching hardware.

* 3.1.1.5 Welders and welding. In addition to the requirements of 3.26.4, all welders employed in the fabrication of the semitrailer shall be certified, before any welding is accomplished, in accordance with MIL-STD-1595; the welding qualifications of the ASME BPVC, Section IX; or AWS B2.1 or D1.1. The certification that the welders have passed the qualifications test shall be on file at the contractor's facility and shall be available for review by the Government.

* 3.1.1.6 Lubrication chart. An illustrated or schematic diagram type lubrication chart shall be provided on the semitrailer. The chart shall direct attention to all lubrication fittings and specify the range and grade of lubricants required for critical temperatures. The chart shall be permanently attached to the semitrailer in a readily visible location. The chart shall be inscribed on a plate conforming to composition A (class 1 or 2) or composition C of MIL-P-514.

3.1.1.7 Wood treatment. Wood shall be treated in accordance with MIL-STD-1223 except interior paneling may be resin impregnated with an overlaid surface on one side of resin treated cellulose fiber and the interior floor may be laminated hardwood in lieu of MIL-STD-1223 treatment.

3.1.1.8 Drain plugs. All drain plugs installed throughout the vehicle for the purpose of draining lubricants shall be of the permanent magnet type.

* 3.1.1.9 24-volt lighting.

* 3.1.1.9.1 Independent 24-volt dc system. When specified (see 6.2), in addition to the standard 12-volt direct current (dc) lighting system, a 24-volt dc lighting system, independent of the 12-volt dc lighting system, shall be provided. Separate wiring and bulbs shall be provided for the 24-volt dc system. The system shall conform to 3.13. The 24-volt dc system shall also conform to the Federal Motor Carrier Safety Regulations referenced in 3.13.1, except the 24-volt dc stop and turn lamps shall be combined.

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* 3.1.1.9.2 Interconnected 24-volt dc system. When specified (see 6.2), capability of operating the normal 12-volt dc system from a towing vehicle equipped with a nominal 24-volt dc electrical system shall be provided.

* 3.1.1.9.3 Receptacle for 24-volt dc system. When either an independent or interconnected 24-volt dc system is specified, the front of the semitrailer shall also be equipped with a 12-contact receptacle and cover conforming to MS 75021, part number MS 75021-1. The receptacle shall be located in accordance with SAE J702. The 24-volt dc, 12-contact receptacle shall be connected to the lights as follows:

Contact B. Connect to left-hand turn signal and stop lamp (yellow)
Contact D. Connect to ground (white)
Contact E. Connect to clearance, side marker, identification, and tail lamps (black and brown)
Contact J. Connect to right-hand turn signal and stop lamp (green).
Contact L. Connect to ground (white).

The remaining contacts shall not be connected. Circuits B and J on tactical (military design) trucks are combination stop and turn indicator circuits. On the interconnected 24-volt dc system, the normal 12-volt dc turn signal lights will function both as turn signals and stop lights, and the normal 12-volt dc stop lights will not be operational when the semitrailer is connected to a towing vehicle with a 24-volt dc power supply. Because of this condition, the stop light (red) circuit is not connected to the 24-volt, 12-contact receptacle.

* 3.1.1.9.4 Resistors. When an interconnected 24-volt dc system is specified, the 12-contact receptacle shall be provided with resistance in each circuit to reduce the voltage of a tactical towing vehicle from a nominal 28 volts dc to 12 volts dc. Each circuit resistor shall be selected to reduce the voltage of the tactical truck to within the maximum rated voltage of the semitrailer electrical components. The resistor assembly shall be located in a protective housing and provided with adequate ventilation or a heat sink to prevent overheating and any heat damage to resistors, wiring, or adjacent components.

* 3.1.1.10 Lifting and tiedown attachments. The semitrailer shall be equipped with lifting and tying down attachments. Lifting and tying down attachments shall conform to type II or type III of MIL-STD-209. A transportation plate conforming to composition A (class 1 or 2) or composition C of MIL-P-514 shall be attached to the semitrailer. The transportation plate shall be inscribed with a diagram showing the lifting attachments and proposed lifting slings, the capacity of lifting attachments and proposed lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the semitrailer, showing the center of gravity, shall be provided on the transportation plate. Tiedown attachments shall be identified by stenciling or other

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suitable markings. Tiedown markings shall clearly indicate that the attachments are intended for the tiedown of the semitrailer on the carrier when shipped.

* 3.1.1.11 Air transportability. The semitrailer, complete with all components and accessories, exclusive of the demountable components specified herein, shall be air transportable in accordance with MIL-STD-1791 and AFSC Design Handbook DH-1-11 in type C-130 aircraft, equipped with the A/A32H-4A Cargo Handling System. Preparation for air transport shall be accomplished by the removal of the running gear assembly and removal or folding up of the landing gear, leveling jacks, spare tire carrier, spare wheel assembly and rear bumper assembly. All fastening devices for these components shall be corrosion-resistant. All fastening devices except common bolts and nuts shall be secured with corrosion-resistant security chains or cables to prevent loss. Spring loaded dowel pins, alinement pins, and other quick release methods of alining removable components shall be provided. Torque requirements for threaded fasteners for installation purposes shall be provided in written instructions. The demountable body shall be capable of being winched to and from aircraft when supported on the body's skid frame. The vehicle shall be designed so that preparation for air transport can be accomplished in not more than a total of two manhours.

3.1.1.11.1 Air transportability verification. Three copies of comprehensive scale drawings satisfactorily showing each stage of loading, tiedown and unloading, for the aircraft specified above, will be required to be furnished to the engineering offices of the procuring activity, TACOM, AMSTA-UED, for subsequent review and approval of responsible Air Force Officials at Headquarters, Aeronautical Systems Division (AFSC), Wright-Patterson Air Force Base.

3.1.1.11.2 Scale drawings. Scale drawings shall include plan, side and end views of the semitrailer and outlines of the semitrailer negotiating required ramps and clearing required overhead door openings. As a minimum, the following information shall be shown:

WEIGHT/CAPACITY
- (a) Weight in shipping configuration
- (b) Number of tiedowns with accompanying capacity and location
- (c) Payload rating.

MISCELLANEOUS
- (a) Vehicle nomenclature and semitrailer's trade name, if any
- (b) Manufacturer's model number
- (c) Manufacturer's name
- (d) Point of contact at the semitrailer manufacturer
- (e) Telephone number at the semitrailer manufacturer
- (f) Critical characteristics of the contractor selected Government truck tractor to be used to load the semitrailer on the aircraft.

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3.1.1.11.3 Air transportability data plate. An air transportability data plate conforming to MIL-P-514, composition A (class 1 or 2) or composition C, shall be provided. The plate shall contain at least the following information:

(a) Side and rear silhouette views of the semitrailer
(b) Horizontal and vertical locations of the center of gravity of the semitrailer in transport condition
(c) Shipping weight
(d) Overall length, width and height
(e) Tiedown instructions for air transport in type C-130 aircraft
(f) Required critical characteristics of the truck tractor to be used to load the semitrailer on the aircraft.

3.1.1.2 Towing devices. Two towing devices shall be provided on the front and two on the rear of the body. The towing devices shall be adequate for pulling the body on its skid into an aircraft and of pulling the body from the aircraft.

3.1.1.13 Lifting pads. Lifting pads shall be provided at the rear end and on each side rearward of the gooseneck or landing gear. The lifting pads shall protect the underside of the semitrailer and shall be capable of accepting their portion of the gross weight when lifting the body with a fork lift truck for removal and installation of the rear axle assembly and the landing gear assembly, if not retractable. Skid rails at the rear of the body shall serve as the rear lifting pads. A minimum size of 200 millimeters (mm) (8 inches) wide by 760 mm (30 inches) long lifting pads shall be installed on each side of the body. The side lifting pads shall be located on the outside edge of the vehicle body 790 mm (31 inches) rearward from the edge of the gooseneck for gooseneck semitrailers or from the landing gear centerline for semitrailers without a gooseneck, measured to the centerline of the lifting pad. The bottom of the lifting pads shall be on the same horizontal plane as the bottom of the skid rails.

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards applicable to the type of semitrailer furnished and in effect on the date of manufacture.

* 3.2.2 OSHA and FMCSR regulations. The vehicle, with all furnished attachments, accessories and equipment, shall enable compliance by any user with all OSHA and FMCSR user regulations that are vehicle or vehicle equipment dependent. Regulations shall be those that are applicable to a commercial user of such a similar type, class and size of semitrailer for the same general use. See 6.1.

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3.2.3 Ratings. Vehicle ratings shall be the manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings must be submitted to the engineering office of the procuring activity.

* 3.2.4 Dissimilar metals. All dissimilar metals used throughout the vehicle shall be insulated from one another to prevent galvanic or electrolytic action.

* 3.2.5 Prohibited materials. Asbestos materials shall not be used in any form in any part of the vehicle. No item, part or assembly shall contain radioactive materials in which the specific activity is greater than 0.002 microcurie per gram or activity per item equals or exceeds 0.01 microcuries.

* 3.2.6 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

* 3.2.7 Lubrication. Lubrication means shall be provided for all parts of the semitrailer normally requiring lubrication. Where the use of high lubricating pressure will damage grease seals or other parts, fittings with pressure release shall be used.

* 3.2.8 Safety. All equipment or exposed portions of the equipment which are subject to extreme temperatures and inclement weather and all rotating or reciprocating parts which are of such a nature or so located as to become a hazard to operating personnel shall be insulated, fully enclosed, or properly guarded.

* 3.2.9 Electromagnetic radiation. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551.

* 3.3 Performance. The semitrailer shall evidence no part failure, deformation, permanent set or interference between parts when towed, both empty and when loaded as specified in 3.4:

(a) At speeds as great as 32 kilometers per hour (km/h) (20 miles per hour (mph)) over unimproved roads and reasonably hard uneven terrain;
(b) At speeds as great as 97 km/h (60 mph) over improved roads; and
(c) At speeds as great as 16 km/h (10 mph) on 10 percent side slopes on unimproved roads and reasonably hard uneven terrain.

3.3.1 Turning ability. The semitrailer shall be capable of assuming an angle of 90 degrees to a coupled truck tractor without cramping or damage to the semitrailer or the truck tractor.

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* 3.3.2 Tracking ability. The semitrailer shall conform to the tracking requirements at Federal Motor Carrier Safety Regulation 393.70(a) when operated under the conditions specified in 3.3.

* 3.3.3 Brake performance. The service brakes shall stop the tractor-semitrailer combination, with the semitrailer loaded as specified in 3.4, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52. Requirements shall be met with the truck tractor loaded to its full rated gross vehicle weight (GVWR) and gross combination weight rating (GCWR).

3.3.4 Air conditioning and heating performance.

3.3.4.1 Air conditioning performance. The air conditioner shall be capable of maintaining not more than 27° Celsius (C°) (80° Fahrenheit (°F)) dry bulb and a 20°C (68°F) wet bulb temperature in the vehicle in an ambient air temperature of 43°C (110°F) under full solar radiation and peak heat load. The worst condition shall be with a 5-kilowatt (kW) peak heat load, four people, and full solar radiation.

3.3.4.2 Heating system performance. The heating unit of the system shall be capable of maintaining a 13°C (55°F) temperature in the vehicle in a -46°C (-50°F) ambient air temperature when the semitrailer is exposed to a 81 km/h (50 mph) wind.

3.3.5 Waterproofness. The vehicle shall be waterproof to prevent entrance of water when exposed to a spray of 0.18 megapascals (mPa) (25 pounds per square inch (psi)) located 610 mm (two feet) from the vehicle for at least 15 minutes. The spray shall be projected on the roof, all sides and the undercarriage of the vehicle.

3.4 Semitrailer weights and payload. The rated payload capacity, evenly distributed in the semitrailer rear compartment load area, shall be not less than 4990 kg (11,000 pounds).

3.4.1 Curb weight. The curb weight shall include the weight of the semitrailer with installed equipment, all attachments and accessories and shall not exceed 6350 kg (14,000 pounds).

* 3.5 Dimensions and clearances. The exterior dimensions and clearances shall be as specified in table I when measured with the vehicle uncoupled from the prime mover, resting on the landing gear and tires, with the interior floor level; when in air transport position; and without payload. The overall length includes the front mounted air conditioner. The overall width excludes the handles, lights, and other appurtenances.

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TABLE I. Dimensions and clearances.

General location	mm	(Inches)
Overall length, +0, -150 mm (+0, -6 inches)	9750	384
Overall width +25, -0 mm (+1 inch, -0)	2410	95
Overall height (maximum)	3560	140
Upper fifth wheel height ± 25 mm (± 1 inch) (distance from ground to underside of fifth wheel plate at centerline of kingpin)	1220	48
Kingpin to front of vehicle ± 75 mm (± 3 inches):	840	33
Swing radius (maximum)	1470	58
Landing wheel clearance (distance from centerline of kingpin to any point on the vehicle 150 mm (6 inches) or more below upper fifth wheel plate), not less than	2080	82
Ground clearance (minimum)	240	9.5
Overall height of vehicle, resting on skid rails in air transport position (maximum)	2590	102

3.6 Design and construction. The vehicle shall have a fully enclosed van type body, with removable chassis components to provide for loading into a C-130 type aircraft. The semitrailer shall have a straight frame with a single level floor, or shall have a gooseneck type frame with a single or two-level floor. When an inside air conditioner is furnished, a transverse partition shall be furnished over the gooseneck or forward of the landing legs. The vehicle shall be equipped with double rear doors and a curbside door adjacent to the landing legs. The vehicle shall be equipped with a vertical heating and air conditioning system, mounted on the floor on the streetside at the forward left corner of the front compartment or mounted outside on the front wall.

3.6.1 Body types. One of the following body types shall be furnished:

(a) Chassis frame
(b) Trussed body side members
(c) *Stressed-skin (monocoque or semi-monocoque).*

3.6.1.1 Body exterior and framing. Body exterior, except the front and the doors, shall be corrugated aluminum alloy sheets having corrugation on 100 mm (4-inch) centers for the side walls or shall be smooth skin, stretcher level aluminum alloy sheets. Body, underbody, and roof framing shall be constructed of steel or aluminum sections. Steel or aluminum structural members shall be arc welded and reinforced at joints to fulfill the dynamic and static loading requirements specified herein. When splices are necessary, they shall be designed to avoid stress concentration. Splice welds shall be continuous. Welding material shall be compatible to the material being welded. Structural members shall be designed to accept the loads imposed by the various operations specified herein. The roof shall be of one-piece construction of aluminum alloy sheet.

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depletion of the compressed air supply. The parking brakes shall be automatically applied upon disconnection of the emergency air line and under emergency conditions. The parking brakes shall conform to Federal Motor Carrier Safety Regulation 393.41.

3.11 Upper fifth wheel plate. The upper fifth wheel plate shall be of the one-piece rigid type, not less than 6.4 mm (0.25 inch) in thickness, reinforced, having a minimum dimension of 1140 mm (45 inches) in length and 910 mm (36 inches) in width, and shall conform to Federal Motor Carrier Safety Regulation 393.70(b). The forward end of the upper fifth wheel plate shall have a minimum 25 mm (1 inch) long lip turned up at 45 degrees or a 25 mm (1 inch) rolled lip, for ease of loading onto the towing vehicle. A 50 mm (2 inch) heat treated, alloy steel kingpin conforming to SAE J700 shall be provided.

* 3.12 Landing gear. The vehicle shall have two vertical lift, telescoping, nonrotating landing legs with two-speed gears and a hand-crank for each landing leg. The landing legs shall be equipped with self-leveling skid pads not less than 250 mm by 250 mm (10 inches by 10 inches) in area. A holder shall be provided for each handcrank. A corrosion-resistant security chain or cable, secured to the landing leg, shall be provided except when permanently attached crank handles are used. The landing gear assembly shall be capable of operating independently from either side of the vehicle for side-to-side leveling. The landing gear assembly and its components, accessories, and attachments shall withstand, without deformation, the combined static and dynamic forces due to the portion of the gross weight sustained, and the additional forces resulting from impact during coupling and uncoupling operations. When placed in travel position, the landing gear assembly shall remain locked. The landing gear assembly shall be protected to prevent the entrance of foreign matter which would impair its functioning or mechanical efficiency. A lubrication fitting shall be furnished. The landing gear assembly shall be fully retractable above the skid rails specified in 3.16.4 or shall be removable for air transport.

* 3.12.1 Landing gear range of adjustment. When the landing gear is in a retracted position, the ground clearance shall be not less than 300 mm (12 inches). The landing gear assembly shall have sufficient vertical travel to vary the height of the upper fifth wheel plate, measured at the vertical centerline of the kingpin, from at least 1120 mm (44 inches) to not less than 1320 mm (52 inches) above ground level.

3.12.2 Leveling jacks. Four manually operated leveling jacks shall be furnished. The leveling jacks shall be retractable or removable for highway and air transport. When the leveling jacks are used, they shall be installed in each extreme corner of the vehicle. The leveling jack contact surfaces at the inside of the vehicle frame member corners shall be adequate to meet the stress requirements specified herein. The leveling jacks shall be capable of leveling the vehicle and shall ensure stability under wind velocities of up

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* 3.9.2 Spare wheel or rim. A spare wheel or rim shall be mounted on a carrier located forward of the axle, readily accessible from the curbside. The carrier shall be readily removable for air transport and shall be bolted to the vehicle or mounted to the removable running gear. Threaded fasteners, when used to secure the spare tire in the carrier, shall be constructed of or plated with corrosion-resistant material. The carrier design shall enable safe removal or mounting of a spare wheel assembly using only tools furnished with the vehicle. The carrier shall enable the safe removal and installation of the spare tire assembly from and to the vehicle and carrier without personnel positioning themselves or any part of their body under the spare tire assembly.

* 3.9.3 Spare tire. An inflated spare tire shall be mounted on the spare wheel or rim. The spare tire shall be of the same size, load range (ply rating), and tread design as the tires furnished on the axles.

3.10 Brakes. Full air, internal expanding type brakes shall be furnished on all wheels of the vehicle. The brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.43, 393.45 through 393.50, and 393.52. Brake linings shall be of nonasbestos material. The brake system shall include:

(a) Piping
(b) Hose connections
(c) Gladhands with cover assembly
(d) Air reservoirs
(e) Automatic slack adjusters
(f) Relay emergency valve
(g) And all other associated parts and connections required for a complete air brake system.

Location of air fittings shall be in accordance with SAE J702. Air fittings shall be mounted in a recessed location. Front air hose connections shall be provided with gladhands, spring loaded dust covers, or dummy gladhands equipped with security chains or cables. Gladhands shall conform to SAE J318. All components of the brake system shall be installed in such a manner as to provide adequate road clearance when traveling over uneven or rough terrain, including objects liable to strike and cause damage to the brake system components. No part of the braking system shall extend below the bottom of the wheel rims. The removable running gear assembly shall be provided with air hoses with air couplings that connect to fixed air couplings with spring loaded dust cover on the vehicle chassis. Disconnect is required for air transport. Fixed dummy couplings shall be provided on the running gear frame for use during air transport.

3.10.1 Parking brakes. The semitrailer shall be equipped with spring or air diaphragm mechanical lock type parking brakes. The parking brakes shall hold the semitrailer, with rated payload on a 10 percent grade despite the

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3.6.2 Design load and stresses. The design load shall be equal to the product of 1.5 times the algebraic sum of the vehicle curb weight minus the unsprung weight of the suspension system plus the rated design payload uniformly distributed over the load area. Expressed as an equation:

Design load = 1.5 (curb weight - unsprung weight + rated design payload).

The 1.5 factor shall include and adjust the effect of a shock load (0.50) in stress calculations. When subjected to the computed design load, the normal stress (variously designated as the principal, bending, flexural, or fiber stress) in the load carrying members shall not exceed 50 percent of the material yield strength. The stress in the compression members shall not exceed the following, whichever is less:

(a) 1/2 of critical stress based on Euler column formula.

$$SCR = \pi^2 E (r/L)^2$$

(b) 2/3 of the material yield strength, or
(c) 1/3 of the material ultimate strength.

* 3.7 Axle. The axle rating shall be at least equal to the load imposed, measured at the ground, with the vehicle loaded as specified in 3.4. The wheel bearings and axle spindles shall be oil lubricated. The oil viscosity shall be in accordance with the manufacturer's recommendation. The hubcaps shall have a window for visual determination of oil level. Provisions for venting or another method of withholding internal pressure buildup and for replenishing the oil supply, shall be incorporated.

3.8 Suspension system. The vehicle shall be equipped with a spring suspension system, with each component having a rated capacity at least equal to the load imposed, measured at the ground, with the vehicle loaded as specified in 3.4. The vertical tire clearance shall be not less than 115 mm (4.5 inches). The suspension and axle shall be removable as a unit assembly and shall be as specified in 3.1.1.11.

3.9 Wheels, rims and tires. The vehicle shall be equipped with dual wheels. Rims and tire ratings shall conform to Tire and Rim Association or the European Tyre and Rim Technical Organisation recommendations for the type and size of tires furnished. Tire size, load range (ply rating), tread design, and manufacturer shall be the same for all wheels on the semitrailer. Rim sizes shall be the same for all wheels on each vehicle. Multi-piece rims shall not be furnished.

* 3.9.1 Tires. Tires shall be tubeless steel belted radial ply type with highway tread. Tires shall be of a rated capacity at least equal to the loads imposed on each tire, measured at each wheel, at the ground, with the vehicle loaded as specified in 3.4. Tires shall conform to Tire and Rim Association Recommendations, or European Tyre and Rim Technical Organisation recommendations.

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to 80 km per hour (50 mph). The leveling jacks shall be Westran Model No. 6439 or equal, having a total expanded length of 1520 mm (60 inches), providing a total lift at the front coupler height of approximately 300 mm (12 inches) above ground level. When a clevis and yoke are used to secure the leveling jacks to the vehicle, the fastening devices shall be secured by a corrosion-resistant security chain or cable to prevent loss of the fasteners. The operating mechanism shall be enclosed to prevent contamination.

* 3.13 Electrical system. The electrical system shall be 12-volt dc. Wiring shall be in accordance with SAE J1292, and Federal Motor Carrier Safety Regulations 393.27, 393.28, 393.29, 393.32 and 393.33. All wiring exposed to possible physical damage shall be protected by use of a heavy wall thermoplastic insulated cable or rigid conduit. Wiring within the body area that is not readily accessible for service shall be in a rigid conduit. The conduit shall be provided with vents. All wiring shall be tagged and color-coded in accordance with SAE J560. The conduit shall be adequately and properly secured to prevent possible physical damage by the employment of chafeproof clamps and grommets.

* 3.13.1 Lighting. An electrical lighting system shall be provided. The lighting system shall conform to Federal Motor Carrier Safety Regulations 393.9, 393.11, 393.20, 393.22, 393.23, 393.25 through 393.29, 393.32 and 393.33. All lights and reflectors shall be mounted in recessed locations or in guarded locations to provide protection from operational hazards. Lights and reflectors shall not be mounted on vertical surfaces of rub rails (unless recessed and fully protected), or mounted on vehicle bumpers.

3.13.2 Lighting receptacle. The vehicle shall be equipped with a lighting receptacle mounted in a recessed or protected location at the front of the vehicle. The receptacle shall be in accordance with SAE J560, with a spring loaded cover assembly.

3.14 Bumper. Rear end protection shall conform to Federal Motor Carrier Safety Regulation 393.86. The vehicle shall be furnished with a folding or demountable type bumper as required for air transport. The bumper shall be furnished with corrosion-resistant security chains or cables and corrosion-resistant alinement pins. Alinement pins are for locating purposes. The bumper may be permanently attached to the bogie assembly and be removable along with the bogie itself.

* 3.15 Rear wheel splash protection. Rear wheels shall have rubber mud flaps at the rear of the rear wheels. Mud flaps may be bolted to the bogie frame. Splash and stone throw protection shall conform to SAE J682. Rubber mud flaps when secured to the vehicle chassis, shall be removable for aircraft transport. A metal strip not less than 3.2 mm (0.125 inch) thick and not less than 25 mm (one inch) wide, extending the entire width of the mud flap, shall be installed to prevent the bolt heads or bolt nuts from

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damaging the mud flap. As an alternate method of attaching the mud flaps, tabs or clips with minimum surface contact dimensions of 25 mm (1 inch) high by 32 mm (1.25 inch) wide by 2.4 mm (0.094 inch) thick shall be furnished at each bolt.

3.16 Underbody. The underbody load carrying longitudinal members shall be one piece, without splices except at the kickup for gooseneck type semitrailers. Tire clearance shall be not less than 115 mm (4.5 inches) and shall not extend into the frame member areas.

3.16.1 Platform. The platform shall be flat with no protrusions or wheelhousings, shall be of the single deck type for semitrailers without a gooseneck and of the two level type for semitrailers with a gooseneck. When a semitrailer with a gooseneck is furnished, the step to the front of the vehicle over the gooseneck shall not exceed a vertical height of 300 mm (12 inches). The floor shall be insulated as specified herein. The U-shaped channel and interior tiedown rings specified in 3.19.2 and 3.19.3 shall be secured so as to minimize the transfer of heat.

3.16.2 Floor pan. A floor pan shall be provided to protect the complete underside of the vehicle. The floor pan shall be not less than 18 gage 1.214 mm (0.0478 inch) stainless steel sheet or 1.3 mm (0.050 inch) aluminum. The floor pan shall rise above the loadspace of the floor level and shall have a flanged perimeter except when the body rails are flanged so as to prevent the entrance of water. The floor pan shall be primed and painted to prevent galvanic and electrolytic action. The floor pan shall be watertight to prevent contamination of the insulation cavity.

3.16.3 Floor. A full length, single level or, for a gooseneck semitrailer a two level subfloor shall be provided. The subfloor support members shall be of nonconductive material, primed and painted to prevent galvanic or electrolytic action; or may be oak batten strips treated in accordance with MIL-STD-1223. Insulation as specified in 3.17.6 shall be provided on top of the floor pan and between the nonconductive material. The insulation cavity shall be covered with exterior AC grade plywood, not less than 3/4 inch in thickness, seven-ply, treated in accordance with MIL-STD-1223, and sealed around the perimeter. The subfloor shall be secured to the nonconductive members. The floor shall be capable of withstanding a vertical load of 975 kilograms per square meter (kg/m^2) (200 pounds per square foot (psf)). The interior tiedown devices and U-shaped channel as specified in 3.19.2 and 3.19.3 shall be flush mounted. The subfloor nonconductive members or oak batten strips shall be secured to the crossmembers by slotted or torx head, flat head, self-tapping (thread cutting), phosphate coated or other corrosion-resistant material machine screws of appropriate size and length conforming to SAE J478, Table 4 for slotted screws and recessed head screws. The interior tiedown devices and the U-shaped channel shall be secured to the main frame crossmembers with corrosion-resistant bolts, washers, lock washers, and nuts. An insulation

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insert shall be placed between the members and the tiedowns and the U-shaped channel to minimize transfer of heat. When oak batten support members are furnished, it will be acceptable to secure the interior tiedown devices and U-shaped channels directly to the oak batten support members using corrosion-resistant wood screws of appropriate size and length. Floor seams shall be filled. The subfloor shall be secured with slotted flat head or torx head screws as specified herein. The tiedown devices and the U-shaped channel shall be installed so that when the floor covering is installed, the top edge of the U-shaped channel and the top surface of the mounting plate of the tiedown devices are flush mounted.

3.16.4 Skid rails. Skid rails shall be furnished, or the lower edge of the floor structure may be used as skid rails to provide a skid surface when components have been removed for air transport. Skid rails shall be not less than 300 mm (12 inches) in width and shall mate with the conveyor rollers on the floor of the aircraft (see 3.1.1.11). Skid rails shall be of one continuous length from the rear of the gooseneck to the rear of the vehicle except for interruption at the wheelwells. Interruption shall be chamfered to prevent damage during loading and unloading. The front and rear of each skid rail shall have a chamfered face of the same transverse width as the skid rail. The chamfered face shall be not less than 27 degrees and shall have a vertical rise measured from the bottom edge of the skid rail of not less than 33 mm (1.312 inches). Chamfered edges shall have smooth surfaces that are tangent to the conveyor rollers of the aircraft. The skid rails shall mate with at least the outside pair of aircraft rollers. Skid rail faces in contact with the aircraft rollers shall have a flat, smooth, uninterrupted surface, so that the entire contact surface of each skid rail is in contact with the rollers on the aircraft. Covered, clad or laminated contact surfaces of the skid rails are not acceptable.

3.17 Body exterior. Body construction shall be of a trussed or monocoque type design. The longitudinal upper rails at the roof level shall be continuous, without splices, extending the full length between the front and rear walls of the vehicle, and designed for attachment of full width roof bows, located on not more than 300 mm (12 inch) centers. All vertical structural members shall be one piece without splices. The corner posts shall be such as to present nominal square, rounded or chamfered corners on the vehicle. The contact face on noncompatible metals shall be protected from corrosion.

3.17.1 Trussed body. Trussed body framing structure shall include at least the following: vertical members, consisting of corner and intermediate posts, shall be inserted between the roof structure and the chassis frame, and employed to combine the roof structure to the chassis frame. The corner and intermediate posts shall be continuous without splice between the roof structure and chassis frame except at door opening areas and shall be spaced on not more than 300 mm (12 inch) centers, except different intervals shall be permitted at door opening regions. The vertical posts may be used to add to the rigidity of the truss.

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3.17.2 Monocoque or semi-monocoque body. The stressed skin monocoque or semi-monocoque body unitized structure shall be constructed to prevent buckling of the skin sheets when the vehicle is operating under the intended service. The monocoque design shall incorporate a rubrail feature in the lower structural rail.

3.17.3 Body attachments. Body attachments shall be fastened together in a manner which shall prevent loosening of bolts, screws and rivets, as well as cracking of welded joints when the vehicle is properly maintained and operated under the intended service conditions. Fasteners shall be compatible to the metal being secured. Self-tapping screws shall not be used in areas subjected to stress and wood shall not be utilized for structural framing.

3.17.4 Body side panels. Corrugated body side panels, when furnished, shall be provided with a flat sheet of proper size to accommodate markings (see 3.1.1.2).

3.17.5 Body roof panels. Body roof panels shall be stretcher leveled and shall be secured to the roof bows to prevent panel vibration. The roof shall be capable of withstanding a static load of 480 kilograms on any square meter (kg/m^2) (30 pounds on any square foot). The roof shall be crowned to permit drainage.

3.17.6 Insulation. Body insulation shall have a "K" factor of not more than 0.12 and a density of not less than 32 kg per cubic meter (m^3) (2.00 pounds per cubic foot). The "K" factor of the insulation shall not increase to more than 0.15 in a four-year period. Insulation shall not support combustion or sustain flame, mold, rot, or vermin; shall not corrode metals, absorb odors, disintegrate under vibration, or settle and leave voids; and shall be essentially odorless. Condensation in the insulation cavity shall not exceed 0.23 kilograms per hour (kg/h) (0.50 pound per hour) (pph). Unless foamed-in-place insulation is used, means shall be provided to drain condensation to the ground. Insulation material shall be homogenous and uniform in density and thermal efficiency. Insulation material shall be capable of flexing more than its containing structure without tendency to crack or rupture. Material performance shall be certified.

3.17.7 Drip moldings. Drip moldings shall be furnished over at least all windows and doors.

3.17.8 Rub rails. Rub rails of sheet steel, not less than 10 gage 3.416 mm (0.1345 inch), or fluted aluminum rub rails not less than 3.2 mm (0.125 inch) thick backed by a minimum 10 gage 3.416 mm (0.1345 inch) steel rub rail shall be provided along the streetside and curbside of the body.

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3.17.9 Double rear doors. Double rear doors shall provide full interior width opening for access into the body rear compartment. Interior door opening height shall be not less than 1830 mm (72 inches). The doors shall be hung between box posts with not less than three heavy duty strap hinges on each door. Doors shall be hung to permit folding back against body sides. Means shall be provided to retain doors in fully opened position. Doors shall be insulated (see 3.17.6). The door clearance shall not exceed 9.5 mm (0.375 inch). Designs requiring a center door post shall not be acceptable.

3.17.10 Side door. A side door on the curbside rearward of the gooseneck or landing legs and as near the gooseneck or landing legs as construction permits shall be furnished. The door shall open towards the front of the vehicle. The door opening shall be not less than 710 mm (28 inches) in width and 1830 mm (72 inches) in height. The door shall be hung to permit the door to fold back against the body side. Means shall be provided to retain the door in a fully opened position. The door shall be insulated (see 3.17.6). The door clearance shall not exceed 9.5 mm (0.375 inch). The door shall be hung between box posts with not less than three heavy duty strap hinges.

3.17.10.1 Side door window. A fixed window shall be installed in the side door, located centrally and approximately 1520 mm (60 inches) above the floor. The window shall be glazed to provide for a 150 mm (6 inch) diameter, shatterproof glass or plexiglass window, 6.4 mm (0.25 inch) in thickness. The window shall be provided with a spring-loaded hinge with a cover on the interior, operable from the interior only. The cover shall have a spring-retention latch. A rubber gasket shall be provided on the perimeter of the cover that faces the interior door surface to assure that light does not pass through.

3.17.11 Rear door hardware. Each door shall lock independently of each other in the closed position. The curbside rear door shall be designed to open from inside or outside the vehicle. Each door shall be provided with cam type locks with top and bottom vertical drawbolts. The curbside rear door shall be provided with interior and exterior handles enabling the closing and opening of the door from inside and outside the vehicle by personnel with arctic gloves, and shall be operable in an ambient air temperature of -46°C (-50°F). The rear curbside door shall be provided with a handle having a provision on the exterior to accommodate locking the door with a padlock. Locking mechanism shall have an internal safety release handle to enable doors to be unlocked from the inside of the van when locked from the outside.

3.17.12 Side door hardware. The side door shall be capable of being locked from the inside or outside of the vehicle. A cam type lock with top and bottom vertical drawbolts located on the interior of the door shall be furnished. The lock shall be recessed within the door. The exterior handle shall be provided with a provision to accommodate locking the door with a

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padlock. Lock mechanism shall have an internal safety release handle to enable the door to be unlocked from the inside of the van when locked from the outside. The door hardware shall be operable by personnel wearing arctic gloves, and shall be operable in an ambient air temperature of minus -46°C (-50°F).

3.17.13 Door gaskets. Doors shall be provided with weatherproof, nonabsorbent, light-tight, rubber sealing gaskets. Gaskets shall be fitted around the door openings and between the doors. Gaskets shall have appropriate flexural and compressibility characteristics and shall not absorb moisture. When doors are closed, light, wind, or rain shall not penetrate. Gaskets shall withstand an ambient air temperature range of -46°C (-50°F) to 71°C (+160°F).

3.17.14 Detachable ladder. A detachable ladder of the flatstep type and of lightweight construction shall be provided for the right rear and side doors for use by personnel in entering and leaving the van body. Ladder steps shall be self-cleaning with an antiskid surface. The ladder shall support a static load of 160 kg (350 pounds) centrally located on any single step without failure. The ladder side members shall be designed to provide not less than 30 degrees slope from the vertical. The steps shall provide approximately equal risers, of not more than 200 mm (8 inches) between the ground and the vehicle floor level, and shall furnish sufficient tread width of not less than 200 mm (8 inches) to minimize tripping hazards. A removable hand rail with provisions for locking in a stationary position shall be furnished for use with the ladder. Detachable ladders shall be provided for the side entrance door and the rear curbside service door. In addition, means may be added to position the ladder in the center of the rear doors. Means shall be provided to retain detachable ladders at the door sills when in use.

3.17.15 Detachable ladder carrier(s). Detachable ladder carrier(s) shall be furnished. Carrier(s) shall be removable for aircraft transport. Carrier(s) shall be located in an accessible location under the body.

3.18 Partition. When the air conditioner is located inside the semitrailer, a partition shall be furnished at the front between the rear and front compartments at the kickup or forward of the landing legs. A sliding door shall be furnished (see 3.18.1), and located transversely 660 mm (26 inches) from the curbside interior wall. The sliding door shall provide a clear opening of 760 mm (30 inches) in width and 1520 mm (60 inches) in height. When an outside front air conditioner is provided, the partition shall not be furnished.

3.18.1 Sliding door hardware. The sliding door hardware shall enable the door to remain in the open position or the closed position during highway or aircraft travel. The track for the sliding door shall provide positive alinement and prevent separation of track sections at the joints. End stops

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with padded faces to prevent vibration shall be furnished. Door roller guides with nylon rollers to prevent side-to-side vibration shall be furnished. Door hangers shall be furnished with double nylon rollers. Door hangers shall be furnished with a mechanical means to adjust door alinement. The door bottom face parallel to the floor shall be provided with nylon roller guides operating in a sliding door track with wipers to prevent contamination of the door sill track. The track shall be flush mounted with the vehicle body floor. A sliding door lock with pillar type locking features shall be furnished. The lock shall be furnished with handles on each side of the door. With the handle installed, the handle shall face down to prevent accidental unlatching of the door lock. Striker plates shall be furnished. Locks shall be of the slam type for sliding doors.

3.19 Vehicle body interior. When a semitrailer with gooseneck is furnished, the vehicle body over the gooseneck shall have a finished floor-to-ceiling height of not less than 1750 mm (69 inches), except with the air dampers and lighting fixtures installed, the height shall be not less than 1650 mm (65 inches). The full length of the trailer with a single level floor, and the length to the rear of the gooseneck for a two-level platform, shall have a floor-to-ceiling height of not less than 2030 mm (80 inches), except that with the air dampers and lighting fixtures installed or mounted within the ceiling, the height shall be not less than 1930 mm (76 inches). Under the duct work, within the first 1520 mm (5 feet) from the front, the height shall be not less than 1220 mm (48 inches). The following components shall be recessed or mounted within the vehicle body walls, ceiling and floor:

(a) U-channel (see 3.19.2)
(b) Interior tiedown rings (see 3.19.3)
(c) Air distribution ducts and dampers (see 3.20.6)
(d) Compressed air system (see 3.21)
(e) Electrical conduit and receptacles (see 3.13).

3.19.1 Interior paneling. The interior walls, ceiling, and partition shall be lined with not less than 1/4-inch exterior grade plywood. Plywood shall be treated as specified in MIL-STD-1223 or the plywood shall be resin impregnated with an overlaid surface on one side of resin treated cellulose fiber. The structural members of the body interior walls and ceiling, except partition, shall have oak batten strips, not less than a nominal 50 mm by 50 mm (2 inches by 2 inches), or nonferrous upright stringers secured to high impact plastic spacers and screws. Space shall be provided for the flush mounted components specified herein. The interior finish paneling shall be secured to the oak batten strip, or the nonferrous uprights, by use of countersunk screws of a length that shall not penetrate through the batten strips.

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3.19.2 U-channel. U-shaped channels having a minimum dimension of 12 gage 2.657 mm (0.1046 inch), by 35 mm (1.38 inches) in depth, 35 mm (1.63 inches) in width, with an opening of 22.2 mm (0.875 inch) shall be furnished. The channels, for reference purposes, shall be model P-3000, manufactured by Unistrut Corporation or equal. The channels shall be the full length of the body and shall be recessed and flush mounted to the finished floor, side walls, and ceiling. The U-shaped channels shall be installed to minimize the transfer of heat at the points of attachment (see 3.17.3). The channels shall be of corrosion-resistant material. An end cap shall be furnished in the end of each row.

3.19.2.1 Channel locations. The U-shaped channels shall be located in two longitudinal rows recessed in the ceiling and in the floor. Channels shall be parallel to each other. Channels shall be measured from the centerline of the channel. The floor and ceiling channels shall be 610 mm (24 inches) from the side walls. Each side wall shall have three longitudinal rows. One row shall be 100 mm (4 inches) from the floor, one row shall be 100 mm (4 inches) from the ceiling, and one row shall be spaced equally between the top and bottom channels.

3.19.3 Interior tiedown rings. Tiedown rings shall be furnished. Tiedown rings, for reference purposes, shall be model 170 or model 172 recessed rope ring, as manufactured by Eberhard Manufacturing Company, or equal. Each tiedown ring and its attachment shall be capable of withstanding a 910 kg (2,000 pound) pull in the vertical position and at a 45 degree angle to the vertical position. Tiedown rings shall be so installed as to minimize transfer of heat and shall be mounted flush with the floor. Tiedown rings shall be corrosion resistant.

3.19.3.1 Tiedown ring locations. Tiedown rings shall be located in two longitudinal rows flush mounted in the floor as specified herein. Tiedown ring locations shall be 300 mm (12 inches) from the longitudinal centerline of the vehicle to the centerline of the ring of the tiedown plate. Tiedown rings shall be longitudinally spaced 610 mm (24 inches) apart on the centerline of the ring. The first tiedown ring in each row shall be located on the first crossmember that is closest to 1830 mm (6 feet) from the rear of the vehicle, with six tiedown rings in each row, and rings shall be perpendicular to each other on the transverse plane, i.e., the opening direction of the tiedown rings in the left and right rows of tiedowns shall be to the left and right side walls of the body respectively.

3.19.4 Floor covering. The complete floor area and any steps over the gooseneck shall be covered with vinyl conforming to L-F-475. The color of the floor covering shall be light tan. The floor covering shall be installed with waterproof cement and shall lay flat with no air pockets. Seams shall be tight to one another. Entrance doors and steps shall have corrosion-resistant, extruded metal moldings to protect the edges of the floor covering. The floor covering shall be sealed around its perimeter to prevent water entrapment and to prevent corrosion.

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3.20 Air conditioner and heater.

3.20.1 Air conditioner. An air conditioner shall be furnished. The air conditioner shall meet the performance requirements specified in 3.3.4.1. The air conditioner shall have a rated capacity of not less than 10 550 watts (W) (36,000 British thermal units per hour (Btu per hr)). The air conditioner shall be located either on the outside front or in the forward compartment in the left front streetside of the vehicle. The air conditioner shall be of vertical design. The air conditioner shall be designed to operate on a 208-volt, 3-phase, 60 hertz (Hz), 4-wire alternating current (ac). The air conditioner shall be mounted and insulated to prevent transfer of vibration under any of the operating conditions specified herein. When an inside mounted air conditioner is furnished, it shall be capable of being removed for service through the partition sliding door. When an outside front air conditioner is furnished, the partition specified in 3.18 shall not be furnished.

3.20.2 Heaters. When the air conditioning system is being used for heating the vehicle, the heating system shall meet the requirements specified in 3.3.4.2.

3.20.3 Controls. Air conditioning and heating controls shall be located near the air return opening and at a point to allow easy reading and adjustment, or they may be combined with the thermostat and located as specified in 3.20.4. Controls shall be capable of changing from cooling to heating.

3.20.4 Thermostat. A thermostat shall be furnished to maintain heating and cooling settings and shall be manually adjustable within $\pm$ 2°C ($\pm$ 4°F). The thermostat shall have an index range from 10°C (50°F) to 32°C (90°F). The thermostat shall be located in the forward section of the van near the air return opening. The thermostat shall not be located near an air outlet.

3.20.5 Condenser air opening. When an inside mounted air conditioner is furnished, the condenser air opening shall be provided in the vehicle wall for flow of air to and from the condenser and to the fresh air inlet. The opening shall be air and watertight between the walls of the vehicle and around the opening. The opening shall have automatically operated louvers with screens.

3.20.6 Air distribution system. An air distribution system shall be furnished which recirculates conditioned air and warmed air through air ducts into the front and rear of the vehicle, and returns the air to the air conditioner and heater unit. Heating and cooling air ducts protruding from the ceiling shall be insulated on the topside and covered with a nonabsorbent type of waterproof covering on either exterior or interior surface areas that protrude below the ceiling. However, ductwork need not be insulated when it

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is located below an insulated ceiling. The covering shall prevent condensation forming on the exterior surfaces. The floor to ceiling height may be reduced by the thickness of the covering. The air conditioning and heating unit shall be provided with a plenum chamber connecting the unit and the external opening (see 3.20.10) with the conditioned air system. A damper shall be furnished for operating the unit. The air distribution system shall be corrosion-resistant. The system shall be so installed as to prevent transfer of vibration.

3.20.7 Conditioned air discharge. The cooled and warmed air distribution system shall extend from the front to the rear of the vehicle. The system shall be provided with dampers in the duct system to regulate the air velocity. Dampers shall indicate the open or closed position. The cooled and warmed air shall discharge through manually operated damper diffusers.

3.20.8 Discharge dampers. Discharge air dampers shall be installed in the ceiling. The dampers shall have adjustable diffusers to regulate the flow of air. The damper diffusers shall direct the air movement toward the sides of the body. Dampers shall be manually operable and shall provide a floor-to-damper height as specified in 3.19.

3.20.9 Pressure drop. The air distribution system shall have a pressure drop of not more than 19 mm (0.75 inch) of water.

3.20.10 External air distribution connection. Two heavy duty, 200 mm (8-inch) outer diameter (od), sheet metal connectors, with a rolled lip on the outer edge covered with an airtight and watertight hinged cover, shall be provided. The external input duct connection shall be connected to the plenum chamber. A damper shall be provided adjacent to the interior wall which shall indicate the opened or closed position. The hinged cover shall have a rubber gasket with a standard, commercial, corrosion- resistant, spring loaded latch. The latch shall be designed to hold the cover closed so that the cover is watertight, rattle-free, and does not come open or loosen during transport of the semitrailer. The external duct connector shall supply warm or cool air from an external source in lieu of the on board air conditioning and heating unit. Connections shall be located near the left front of the vehicle.

3.21 Compressed air supply system. A compressed air supply system shall be furnished for receiving and distributing compressed air from a Government supplied external source. This source will be connected to the compressed air system and receiver tank of the semitrailer by means of an external air intake coupling located in the 200 mm (8-inch) by 200 mm (8-inch) opening described in 3.21.1. The system shall conform generally to figure 1 and shall include a compressed air intake coupler, 6 mm (1/4 inch) air lines, air receiver tank, valves, pressure gages, strainer (filter), and 6 mm (1/4-inch) compressed air outlets in the front equipment area and rear working area.

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The air supply lines to the compressed air outlets on the curbside and streetside of the vehicle shall be recessed within the vehicle body walls and suspended to prevent vibration and chafing. The air supply system shall be accessible for service without loss of insulation. Metals shall be compatible to one another to prevent galvanic or electrolytic action.

3.21.1 Compressed air intake. The vehicle shall be furnished with a compressed air intake located in the front or curbside wall of the semitrailer as near the front curbside corner as practical. The air intake coupler connection shall be located in a 200 mm (8-inch) by 200 mm (8-inch) compartment, located not less than 300 mm (12-inches) above floor level, and of sufficient depth for the air line coupler. The compartment shall be furnished with a hinged external door with a rubber gasket and a standard commercial spring loaded latch designed to hold the cover closed, so that the cover is weathertight and will not open or work loose during transport of the semitrailer. The compartment shall be lined with a nonferrous sleeve and shall be airtight and watertight.

3.21.1.1 Air line coupler. A 1/2-inch quick disconnect air fitting shall be furnished in the 200 mm (8-inch) by 200 mm (8-inch) compartment. The fitting shall be of the plug-in type which opens when plugged in and closes when pulled out, and shall be provided with both male and female fittings. The coupler shall be mounted to 3/4-inch seamless, heavy duty, copper tubing, projecting into the interior walls. The copper tubing shall have a working pressure of not less than 2.8 mPa (400 psi) and a wall thickness of not less than 1.7 mm (0.065 inch).

3.21.2 Coupler to receiver tank. The air supply from the coupler to the receiver tank shall be through the 19 mm (3/4-inch) heavy duty, seamless copper tubing. This air line shall be furnished with a 19 mm (3/4-inch) brass air shutoff valve.

3.21.3 Air receiver. The air receiver tank shall be a commercial tank adequate to meet specified requirements and shall have a nominal capacity of 114 liters (L) (30 gallons). The tank shall be of the horizontal type with sufficient capacity to provide 1.4 mPa (200 psi) pressure to the 1/4-inch outlets specified in 3.21.7. The tank shall be constructed in accordance with the ASME Boiler and Pressure Vessel Code, Section VIII. Certification shall be furnished that the receiver tank meets the requirements specified herein. The tank shall be mounted on a substantial bedplate. The bedplate shall be secured to the platform with corrosion-resistant bolts, nuts, washers, and lock washers. An air strainer, with water trap and water reservoir to collect moisture, shall be furnished. The air strainer shall be adjacent to either the air shutoff valve in the air receiver inlet line or in the air receiver discharge line. The water reservoir shall be furnished with an exhaust valve and a line to expel water through the floor to the vehicle exterior. Felt or sintered metal filters and a screen to prevent air line contamination shall be provided.

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3.21.3.1 Air receiver components. Air receiver components shall include at least the following:

(a) Receiver safety and relief valve(s)
(b) Drain valve and line to expel water through the floor
(c) Service valve
(d) Shockproof pressure gage

3.21.4 Receiver discharge air line. The air receiver discharge shall be connected to a 19 mm (3/4-inch) air supply line. The connection shall include a 19 mm (3/4-inch) brass union and a 19 mm (3/4-inch) brass air shutoff valve.

3.21.5 Air supply lines. Air supply lines shall be furnished to each side of the vehicle. The main air line shall be 19 mm (3/4-inch), heavy duty, copper tubing having a working pressure of not less than 2.8 mPa (400 psi) and a wall thickness of not less than 1.7 mm (0.065 inch). The air line shall be connected to at least 16 mm (5/8-inch) service air lines with a union and a 19 mm (3/4-inch) brass valve on each side of the vehicle to permit personnel to shut off the air supply on either side of the vehicle. The 16 mm (5/8-inch) service air lines shall extend from near the receiver tank at the front of the vehicle to within 1220 mm (4 feet) of the rear service doors on each side of the vehicle. The service air lines shall be located 1220 ± 50 mm (48 ± 2 inches) above floor level and shall be recessed into the body wall. The air lines on each side of the vehicle shall be provided with at least a 16 mm (5/8-inch) air pressure regulator as specified in 3.21.6. The air supply lines shall be secured to prevent vibration within the wall and shall be accessible for service. Clamps, when employed, shall be chafeproof.

3.21.6 Air pressure regulators. An air pressure regulator with gage shall be provided for each of the minimum 16 mm (5/8-inch) service supply lines on the streetside and the curbside of the semitrailer. The air pressure regulators shall be adjustable to operate at 1.05 mPa (150 psi) to 0.07 mPa (10 psi). The air pressure gage shall be shockproof against both vibration and line pressure shock. The gages shall indicate air pressure to the service lines on each side of the vehicle. An air pressure regulator shall also be supplied in the 19 mm (3/4-inch) main air supply line, if required to maintain constant pressure in the system.

3.21.7 Compressed air outlets. The service air lines shall be connected to the 6 mm (1/4-inch), wall mounted compressed air outlets. Not less than four 6 mm (1/4-inch) compressed air outlets shall be furnished on each side to the rear of the gooseneck, landing legs, or partition, located 1220 ± 50 mm (48 ± 2 inches) from floor level and 300 ± 50 mm (12 ± 2 inches) forward of each duplex electrical convenience receptacle (see 3.22.2.3.1 and 3.22.2.3.2). One 6 mm (1/4-inch) compressed air outlet shall be furnished in the front equipment area, on the streetside wall above the air receiver

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tank. Each compressed air outlet shall be furnished with a 6 mm (1/4-inch) swivel adapter and a check unit for quick disconnect of a hose line. Outlets shall be mounted as flush to the wall as practicable while still facilitating the use of the swivel adapter quick disconnects.

3.22 Interior electrical system. The vehicle shall be furnished with three electrical systems as follows:

(a) 12-volt dc system (see 3.13. and 3.22.1)
(b) 110-volt ac, single-phase, 60 Hz system
(c) 208-volt ac, three-phase, 60 Hz, four-wire system.

The vehicle interior electrical system shall be in E. M. T. (thinwall) conduit (galvanized electrical metallic tubing) with compression type connectors. Joints shall be threaded with connections at all lamps and junction boxes, excluding boxes around the lights. Junction boxes shall be accessible and shall be provided with junction box covers. Wiring shall be color-coded at the junction boxes and at the distribution panel(s). The conduit shall be recessed within the vehicle body wall, supported with locktight conduit clamps having chafeproof faces and so installed as to prevent vibration. All electrical assemblies, wiring, components and parts shall be weatherproof and shall meet the operating conditions specified herein. Conduit shall be provided with vents in a weather-protected location except when electrical insulation is of the type that is fungus and moisture resistant. Wiring shall conform to SAE J1292 and shall be of sufficient size and gage to accept the loads imposed in operating the electrical system without loss of dielectric strength. The 110-volt ac and 208-volt ac systems shall be capable of receiving power from an outside source. Wiring shall be accessible for service. The electrical system shall be in accordance with NFPA No. 70.

3.22.1 Interior lamps, 12-volt. Two 12-volt ceiling lamps shall be provided. The lamps shall be provided with white nonglare reflectors. One lamp shall be located over the curbside entrance door and one lamp shall be located to provide light to the 110/220-volt power distribution panel(s). Both lamps shall be serviced with a single on-and-off switch, located at the curbside door.

3.22.2 Electrical system, 110-volt. The vehicle shall be equipped with a 110-volt, single-phase, 60 Hz electrical system for interior lighting and for the electrical duplex outlets.

3.22.2.1 Distribution panel, 110-volt. A distribution panel for the 110-volt electrical system shall be furnished. The panel shall be rated for not less than 100 amperes. The panel shall be equipped with automatic circuit breakers and shall be grounded. The panel shall be provided with not less than 10 circuits. In addition, two 15-ampere circuits shall be furnished for future expansion purposes. The panel door shall be hinged and provided with a combination latch, capable of being locked with a padlock.

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The distribution panel shall be furnished with a main breaker or throw switch. A circuit diagram shall be furnished in addition to that specified in 3.1.1.2.1. The 110-volt distribution panel may be combined with the 208-volt, three-phase panel, providing specified circuit breaker protection is furnished for all circuits.

3.22.2.1.1 Distribution panel mountings. The 110-volt or combination 110/208-volt electrical distribution panel shall be located on the interior as close as practicable to the curbside front corner.

3.22.2.2 Electrical lighting and switches, 110-volt.

3.22.2.2.1 Lighting, fluorescent. Fluorescent lamps and fixtures shall be furnished. The lamps shall be located in two longitudinal rows, spaced 760 mm (30 inches) from the curbside and streetside walls. There shall be not less than five double lamp fixtures per row. Each lamp shall be 1220 mm (48 inches) in length and of 40-watt rating. The lamps shall provide a broad distribution of light. Lamps shall be white. A nonglare reflector and tube guard shall be provided. Lamps shall be so installed as to provide a headroom of not less than 1930 mm (76 inches), except for the frontmost lamps when a gooseneck is furnished.

3.22.2.2.2 Lamp switches. Two light switches shall be furnished for the fluorescent lights. One switch shall operate the six fluorescent lights (located in two rows) at the rear of the vehicle and the other switch shall operate the four fluorescent lights (located in two rows) at the front of the vehicle. The switches shall be flush mounted on the curbside partition wall facing the rear compartment, or adjacent to the curbside door. Switches shall be grounded.

3.22.2.3 Electrical wall receptacles, 110-volt.

* 3.22.2.3.1 Electrical wall receptacle (rear working area). Not less than four electrical duplex wall receptacles, with ground fault circuit interrupter protection, shall be furnished on both the curbside and streetside of the vehicle. The receptacles shall be flush mounted. Receptacles shall be located 1220 ± 50 mm (48 ± 2 inches) from the finished floor, with the first curbside receptacle being 910 mm (3 feet) rearward of the curbside entrance door. The first streetside receptacle shall be located 1220 mm (4 feet) rearward of the gooseneck step, landing legs, or partition, whichever is the most rearward. The last receptacle on each side shall be located 910 mm (3 feet) forward of the rear service doors. The intermediate receptacles shall be equally spaced. Receptacles shall be grounded.

* 3.22.2.3.2 Electrical wall receptacle (front equipment area). An electrical duplex wall receptacle with ground fault circuit interrupter protection shall be furnished. The receptacle shall be flush mounted and shall be located 1220 ± 50 mm (48 ± 2 inches) above the finished floor and adjacent to the 110-volt distribution box. The receptacle shall be grounded.

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3.22.2.4 Circuits, 110-volt. The electrical circuits for the lighting and receptacles shall be grouped as follows:

(a) Fluorescent lamp, two circuits of not less than 20 amperes each (one circuit for each row of lamps)
(b) Duplex wall receptacles, two circuits of not less than 15 amperes for each side
(c) Duplex receptacle and ceiling lighting fixtures over the gooseneck (or forward of the landing legs) and entrance light, one circuit of not less than 20 amperes.

The wall switches shall have a capacity of not less than the ampere rating of the circuits.

3.22.2.5 Receptacle face plates. Receptacle face plates shall be of nonconductive and noncombustible material. Each wall switch shall be marked for its purpose and each duplex receptacle marked for its nominal voltage rating (e.g. "110-volts").

3.22.3 Distribution panel, 208-volt. A distribution panel for the 208-volt ac, three-phase, 60 Hz, four-wire system shall be furnished. The distribution panel shall provide 208-volt ac, three-phase, 60 Hz to the air conditioner and heater and shall also include two additional circuits for future expansion. In addition, the distribution panel shall provide 110-volt ac, single-phase, 60 Hz power to the 110-volt distribution panel. The 208-volt distribution panel may be combined with the 110-volt distribution panel. The panel shall be capable of receiving power from an external source. Automatic circuit breakers shall be furnished. The panel shall have sufficient capacity to meet the electrical requirements of the vehicle specified herein with an allowance for the two additional circuits. The panel shall be grounded. The panel door shall be hinged and provided with a combination latch and shall be capable of being locked with a padlock. A throw switch shall be furnished. The circuit diagram shall be furnished in addition to that specified in 3.1.1.2.1.

3.22.3.1 Panel mounting. The 208-volt panel shall be combined with or mounted adjacent to the 110-volt panel.

3.22.4 Power supply receptacle. A 25-kW three-phase, four-wire, 208- to 220-volt receptacle shall be furnished. The receptacle shall be mounted on the front of the semitrailer or flush mounted on the exterior side, on the curbside of the vehicle below the 208-volt distribution panel. The receptacle shall be furnished with a screw cover (threaded) attached with a corrosion-resistant chain and shall be waterproof. The receptacle shall be grounded to the vehicle chassis. The receptacle shall have sufficient dielectric capacity to meet the imposed loads.

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3.22.4.1 Power supply cable. A power supply cable with neoprene rubber cover, not less than 7600 mm (25 feet) in length, shall be furnished. The cable shall have a plug and cap with corrosion-resistant chain on one end and a polarized twist lock fastener and mating connector (for installation by the using activity) on the opposite end. The cable shall be of sufficient dielectric strength to meet the loads imposed. The power supply cable shall be for a 208 to 220-volt, 60 Hz, three-phase, four-wire system.

* 3.23 Fire extinguishers. Two fire extinguishers shall be installed inside the vehicle. The fire extinguishers shall be in accordance with National Fire Protection Association No. 10. The extinguishers shall be type I, class 1, size 20 of A-A-393. Fire extinguishers shall be mounted on brackets with cam type clamps and shall be readily accessible. One extinguisher shall be located adjacent to the curbside door and one extinguisher shall be located adjacent to the rear doors.

3.24 Tools. Special tools that are required for disassembly and assembly of components for aircraft transport shall be provided. Tool stowage shall be provided in the vehicle in the front equipment area. The stowage compartment shall be furnished with a hinged cover, a hasp for a 19 mm (0.75 inch) padlock and a drain hole. The stowage compartment shall have bins with retaining clamps to retain tools during aircraft and road travel. The stowage compartment shall be of not less than 16 gage 1.519 mm (0.0598 inch) sheet steel. The compartment shall be reinforced. The stowage compartment shall be demountable.

* 3.25 Servicing and adjusting. Prior to acceptance of the semitrailer by the Government inspector, the contractor shall service and adjust the semitrailer for immediate operational use. The servicing and adjustment shall include at least the following:

(a) Inflation of all tires
(b) Adjustment of brakes
(c) A check for proper functioning of all lights
(d) Complete lubrication with grades of lubricants recommended for ambient air temperature at the delivery point.

* 3.26 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Bolted, riveted and welded construction used shall be in accordance with the highest standards of the industry.

* 3.26.1 Metal fabrication. Metal used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause damage to the metal. Shearing and chipping shall be done neatly and accurately. All bends of a major character shall be made with controlled means in order to ensure uniformity of size and shape.

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* 3.26.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

* 3.26.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

* 3.26.4 elding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. See 3.1.1.5.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility for ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to

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determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

4.3 First vehicle preliminary inspection. The first in-process vehicle inspection shall be performed by the Government to evaluate workmanship and to determine conformance with contractual requirements.

4.3.1 Installation of recessed components. The first in-process vehicle shall be inspected to verify that the following recessed components are furnished and installed as follows:

(a) Insulation
(b) Partition if air conditioner is mounted inside
(c) U-channel
(d) Interior tiedown ring(s)
(e) Air distribution system
(f) Compressed air supply system
(g) Interior electrical system.

4.3.2 Water spray test. Each vehicle shall be subjected to a 100 percent water spray test before the installation of the insulation and subfloor. The test shall be as specified in 4.4.6.

4.3.3 In-process inspection of vehicle and components. The first completely assembled vehicle and its components shall receive an in-process inspection by the Government prior to installation of the vinyl or linoleum and the prime painting of the interior and exterior of the vehicle. The inspection shall be conducted to evaluate workmanship and to determine conformance to contractual requirements.

4.4 First production vehicle inspection. The first vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance with the requirements of the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.4.1 Weight and payload capacity. The first production vehicle shall be weighed to determine the curb weight and the distribution of the curb weight on the axle and kingpin. The total imposed loading on the axle and kingpin shall be computed by the contractor and verified by the Government using the curb weight and payload distributed as specified in 3.4. Calculated imposed loads on the axle shall be compared to the suspension, axle and tire load capacity ratings to determine if these components are of adequate capacity to meet contractual requirements.

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* 4.4.2 Road test. The first vehicle for each contract line item number (CLIN) under contract shall be road tested with and without payload. The road test shall be for the distance required in table II, corresponding to the number of vehicles being procured under the specific contract line item number (CLIN). The test shall be conducted by the contractor under the direction of Government representatives. The test shall be conducted over a test course approved by the procuring activity and meeting the requirements of table III. The vehicle shall be inspected periodically during and at the completion of the test by the contractor.

* TABLE II. Road test distance.

Number of vehicles being procured	Road test distance km	(miles)
1 - 10	160	(100)
11 - 20	320	(200)
21 - 30	480	(300)
31 - 40	640	(400)
41 - 50	800	(500)
51 - 100	1600	(1000)
101 - up	8050	(5000)

TABLE III. Test course requirements.

Type of road	Road test distance
Dirt	10 percent
Gravel	30 percent
Improved pavement	60 percent

* 4.4.2.1 Test course. The total test mileage shall be on roads and highways within the general geographic limits of the place of manufacture and adjoining states. Any special highway permits required shall be obtained by the contractor. The course shall be arranged to return the vehicle to the manufacturer's plant at the completion of each 1600 km (1,000 miles), if applicable, or at the completion of the test otherwise, for inspection and chassis servicing.

* 4.4.2.2 Test loads. The semitrailer shall be tested for 90 percent of the mileage with the semitrailer loaded as specified in 3.4 and for 10 percent of the mileage with the semitrailer empty. Average speeds during all road tests shall be as close as possible to those specified in 3.3.

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* 4.4.2.3 Road test failure. The vehicle shall successfully complete the entire test. Failure of the vehicle to successfully complete the test shall be cause for nonacceptance of any of the contract quantity, pending correction of deficiencies and evidence of corrective action which shall prevent the recurrence of similar deficiencies in the contract quantity. Failure of the vehicle to successfully complete the test shall not constitute an excusable delay in meeting scheduled deliveries. The equipment shall be closely monitored throughout the test for interference with towing operations, and for instability as evidenced by poor tracking or uncontrollable sideslipping, skidding, swerving, or tilting. The equipment also shall be closely monitored for loosening of parts, interference between parts, leakage of fluids or lubricants, overheating of components, and for damage, permanent distortion, or excessive wear of parts and components. The equipment shall be disassembled to the extent necessary for inspection for unusual wear or damage to components. Substitution of a new semitrailer or replacement of a major component may require complete retest at the discretion of the Government representatives. Rejection of the test semitrailer shall be for damage or deficiencies, including but not limited to the following:

(a) Damage caused by collision
(b) Failure of any major component
(c) Vibration due to misalinement of wheels or frame
(d) Vibration due to the type of construction or mounting
(e) Evidence of abnormal tire wear due to misalinement or unbalance
(f) Failure of any vehicular safety device such as brakes or electrical circuits
(g) Evidence of structural weakness in any part of the vehicles, vehicle components, accessories or welds
(h) Loose mountings of parts or accessories due to workmanship or vehicular operation.

* 4.4.2.4 Brake tests. Brake tests to determine conformance to requirements shall be conducted during the road test. The brake tests shall be conducted with the fully loaded semitrailer coupled to a fully loaded truck tractor. The truck tractor shall have a gross vehicle weight rating (GVWR) of not more than 9070 kg (20,000 pounds) and a gross combination weight rating (GCWR) of not more than 15 850 kg (35,000 pounds). The truck tractor shall be without excessive brake capacity. GVWR and GCWR shall be the maximum ratings on the tractor only and not necessarily the actual maximum GVW and GCW of the combination of test vehicles. Failure of the vehicle combination to stop within the required 12 200 mm (40 feet) from a speed of 32 km/h (20 mph) shall be cause for rejection.

* 4.4.2.5 Turning ability test. A truck tractor shall be coupled to the semitrailer and driven through turns up to 90 degrees to the towed vehicle, right and left. At the conclusion of five turns in each direction, the tractor and the semitrailer shall be examined for evidence of binding, misalinement or other damage.

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* 4.4.2.6 Parking brake test. To test parking brakes, the semitrailer, with rated payload, shall be placed on a grade, first headed up the grade and then headed down the grade, and tested to determine conformance to 3.10.1. These tests shall be conducted with the semitrailer disconnected from the towing vehicle and resting on the landing legs.

* 4.4.3 Landing leg test. With the semitrailer disconnected from the towing vehicle and loaded as specified in 3.4, the landing legs shall be raised and lowered to vary the height of the upper fifth wheel over the landing gear's full range, to determine conformance to 3.12 and 3.12.1. The loaded semitrailer shall be coupled and uncoupled no less than 5 times and the landing legs shall be examined for binding and or misalinement.

* 4.4.4 Electrical test. An electrical test shall be conducted to determine that all vehicle lights are functioning. In addition, when an innerconnected 24-volt system is furnished, the circuits of the 12-contact receptacle shall be energized with not less than 28 volts dc regulated voltage for a time sufficient to stabilize the temperature of the resistor assembly for not less than 30 minutes in an ambient temperature of not less than 25°C (77°F). After the test, the resistor assembly, including resistors, wiring and adjacent components shall be examined for evidence of overheating, deterioration or damage.

* 4.4.5 Lifting and tiedown test. Lifting and tiedown provisions, markings, and the connecting structural members of the vehicle shall be inspected for conformance with MIL-STD-209. All lifting and tiedown provisions and the connecting structural members shall meet the criteria of MIL-STD-209. Test results that show the provisions do not meet the criteria of MIL-STD-209 shall constitute failure of this test.

* 4.4.6 Leak test. The semitrailer shall be subjected to a leak test by spraying fresh water at a minimum rate of 50 mm (2 inches) per hour at any angle between 30° and 45° from vertical. The test shall be performed with the water directed against all four sides, the roof and the undercarriage of the semitrailer simultaneously or section by section and shall continue for not less than 15 minutes (per section). The test shall be conducted prior to the installation of the interior insulation and lining. Following the test the semitrailer shall be inspected for water leakage.

* 4.4.7 Air conditioning. The inside and outside of the vehicle shall be exposed to a 43°C (110°F) ambient air temperature for a minimum of four hours with all doors open. The vehicle doors shall be closed when the inside and outside temperatures have equalized. The air conditioner shall be operated and within 30 minutes the recorded temperature inside the semitrailer shall be not greater than 27°C (80°F) dry bulb and 20°C (68°F) wet bulb.

* 4.4.8 Heating. Heating capability shall be demonstrated to determine conformance to 3.3.4.2.

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* 4.4.9 Rear axle and suspension system. The rear axle and suspension system shall be removed from the vehicle as for aircraft transport. Removal and replacement of the rear axle and suspension system shall be demonstrated by use of the lifting pads. The system shall be easily removable and replaceable in correct alinement positions by the use of dowel pins, and shall be secured with fasteners. Connection of the air brake system with the detachable air brake hose connections specified shall be demonstrated.

* 4.4.10 Compressed air supply system. The compressed air supply system shall be checked for conformance to specified requirements.

4.4.11 Leveling jacks. Leveling jacks shall be checked for conformance to requirements.

4.4.12 Wood treatment certification. Manufacturer's records shall be available to verify that all wood requiring treatment in accordance with MIL-STD-1223 has been so treated.

* 4.4.13 Air transportability verification. The vehicle shall be inspected to determine that it conforms to the contractor's air transportability drawings and data as submitted to the Government for transportability approval. As a minimum, the following angles, dimensions and descriptions shall be checked against the Government approved contractor's drawings and data:

(a) Angle of approach
(b) Angle of departure
(c) Height, longitudinal location and identification of highest component on semitrailer
(d) Dimensions and locations of any significant projections on semitrailer
(e) Curb weight.

* 4.5 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and shall be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in an as new condition for the duration of the contract. Release of the first vehicle for shipment prior to completion of the contract shall be at the discretion of the local Government Quality Assurance Representative (QAR).

* 4.6 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under the contract until corrective action has been taken.

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* 4.7 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting the performance requirements contained herein.

5. PACKAGING

5.1 Vehicle processing. Semitrailers shall be processed for shipment, from the manufacturer's plant to the initial receiving activity, in accordance with the manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The vehicles covered by this specification are intended for use by the Government by transporting the semitrailer by air.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
* (b) Issues of DODISS to be cited in the solicitation and if required, the specific issues of individual documents referenced (see 2.1.1 and 2.2).
(c) Identification of appropriate military service for exterior color and marking (see 3.1.1.1 and 3.1.1.2).
(d) Independent 24-volt dc system, if required (see 3.1.1.9.1).
(e) Interconnected 24-volt dc system, if required (see 3.1.1.9.2).

6.3 Subject term (key word) listing.

Air transportable semitrailer van
National Electrical Code
Welder's qualifications.

6.4 Identification of changes. The margins of this specification are marked with asterisks (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document regardless of the marginal notations and relationship to the previous issue.

Custodians:
Army - AT
Air Force - 99

Review activities:
Air Force - 84

Preparing activity:
Army - AT

Project No.2330-A136

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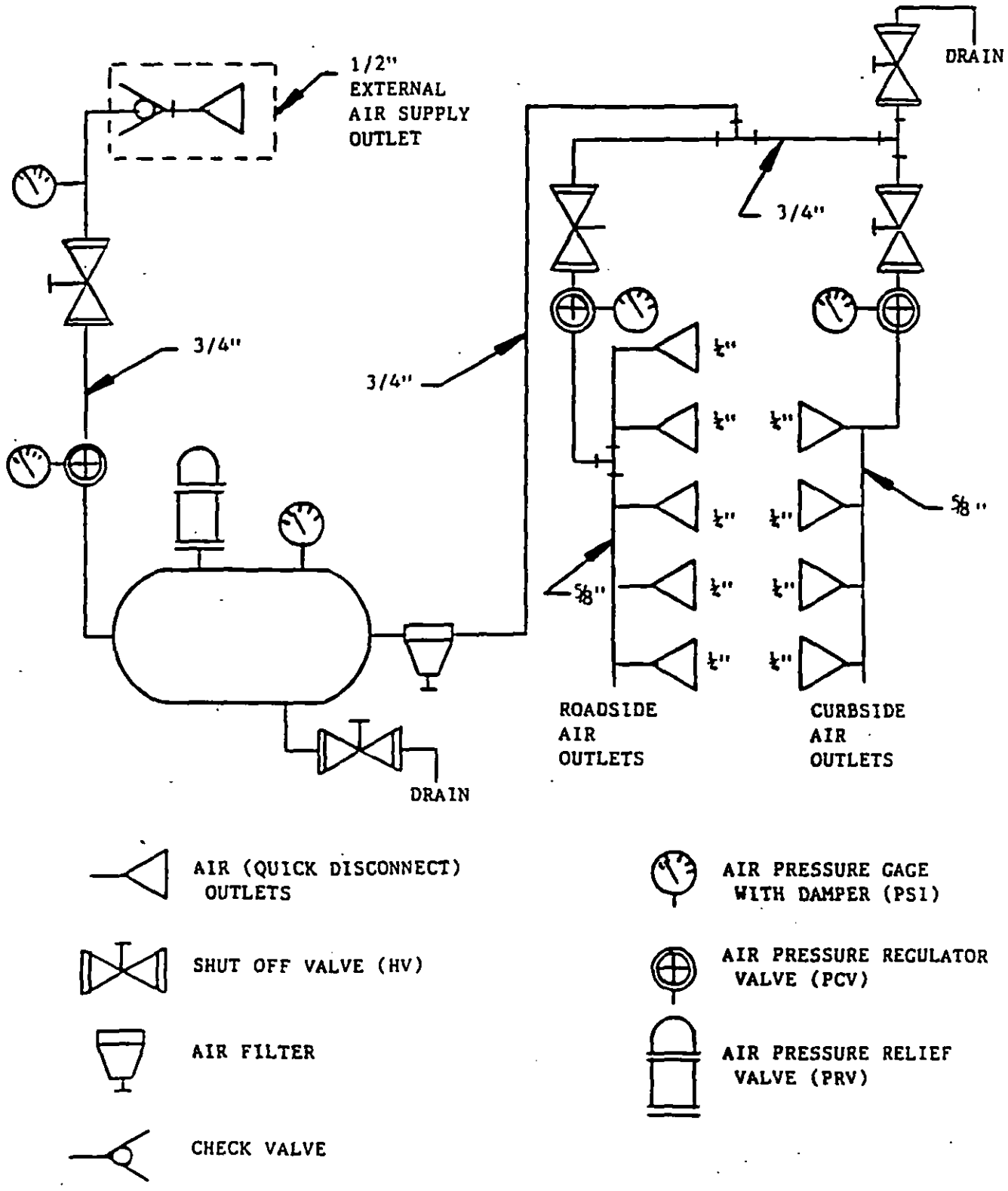


Figure 1 Compressed Air System Typical Schematic

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-62052D	2. DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE SEMITRAILER, VAN; AIR TRANSPORTABLE, 4990 KILOGRAMS (5.5 TON), AIR CONDITIONED, MODIFIED COMMERCIAL

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial: 313-574-6407 (2) AUTOVON: 786-6407
c. ADDRESS (Include Zip Code) US Army Tank Automotive Command Warren, Michigan 48397-5000 AMSTA-MD	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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